Gradient Descent and Optimization

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

2 **[Foundational]** • **Target: Gradient Descent & Optimization Practice Exercises**

[The Multivariable Chain Rule, Directional Derivatives & Gradient Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

3 **[Foundational]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Multivariable Optimization: Critical Points, Hessians & Lagrange Multipliers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

4 **[Foundational]** • **Target: Gradient Descent & Optimization Practice Exercises**

Use the method of Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 25$.

Step 1: Set up the Lagrange equations: Constraint $g(x, y) = x^2 + y^2 = 25$.
$ablaf = \lambda ablag \Rightarrow \langle 3, 4\rangle = \lambda\langle 2x, 2y\rangle$. This gives: $3 = 2\lambda x \Rightarrow x = \frac{3}{2\lambda}$, $\quad 4 = 2\lambda y \Rightarrow y = \frac{2}{\lambda}$. Step 2: Substitute into constraint: $\left(\frac{3}{2\lambda}\right)^2 + \left(\frac{2}{\lambda}\right)^2 = 25 \Rightarrow \frac{9}{4\lambda^2} + \frac{16}{4\lambda^2} = 25 \Rightarrow \frac{25}{4\lambda^2} = 25$. .

5 **[Foundational]** • **Target: Gradient Descent & Optimization Practice Exercises**

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

Set up Lagrange system $ablaf = \lambda ablag$: $ablaf = \langle 2x, 4y\rangle$, $ablag = \langle 2x, 2y\rangle$. 1) $2x = 2\lambda x \Rightarrow 2x(1 - \lambda) = 0$ 2) $4y = 2\lambda y \Rightarrow 2y(2 - \lambda) = 0$ 3) $x^2 + y^2 = 1$ From (1), either $x = 0$ or $\lambda = 1$: • If $x = 0$: from constraint (3), $y^2 = 1 \Rightarrow y = \pm 1$. Points: $(0, 1)$ and $(0, -1)$. Value: $f(0, \pm 1) = 0 + 2(1)^2 = 2$. • If $\lambda = 1$: from (2), $2y(2 - 1) = 2y = 0 \Rightarrow y = 0$. From constraint (3).

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6 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Let $z = x^2y - y^3$, where $x(t) = \cos(2t)$ and $y(t) = \sin(2t)$. Use the multivariable chain rule to find $\frac{dz}{dt}$ when $t = 0$.

Step 1: Compute partial derivatives of z: $\frac{\partial z}{\partial x} = 2xy$, $\frac{\partial z}{\partial y} = x^2 - 3y^2$. Step 2: Compute derivatives of coordinate functions: $\frac{dx}{dt} = -2\sin(2t)$, $\frac{dy}{dt} = 2\cos(2t)$. Step 3: Evaluate at $t = 0$: $x(0) = \cos(0) = 1$, $y(0) = \sin(0) = 0$. $\frac{\partial z}{\partial x}(1, 0) = 0$, $\frac{\partial z}{\partial y}(1, 0) = 1^2 - 0 = 1$. $\frac{dx}{dt}(0) = 0$, $\frac{dy}{dt}(0) = 2$. Step 4: Apply the chain rule.

7 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

Compute gradient: $ablaf = \langle 6xy, 3x^2 - 3y^2 \rangle$. At $(1, 2)$: $ablaf(1, 2) = \langle 6(1)(2), 3(1)^2 - 3(2)^2 \rangle = \langle 12, 3 - 12 \rangle = \langle 12, -9 \rangle$. Maximum rate of change is the gradient magnitude: $|ablaf(1, 2)| = \sqrt{12^2 + (-9)^2} = \sqrt{144 + 81} = \sqrt{225} = 15$ The direction of maximum increase is the normalized gradient: $\mathbf{u} = \frac{ablaf}{|ablaf|} = \frac{1}{15}\langle 12, -9 \rangle = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

Compute partial derivatives via Chain Rule: $f_x = \frac{2x}{x^2 + 3y^2}$, $f_y = \frac{6y}{x^2 + 3y^2}$ Substitute $x = 2$ and $y = 1$: $x^2 + 3y^2 = 2^2 + 3(1^2) = 4 + 3 = 7$ $f_x(2, 1) = \frac{2(2)}{7} = \frac{4}{7}$, $f_y(2, 1) = \frac{6(1)}{7} = \frac{6}{7}$

9 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

10 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

11 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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12 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

13 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

14 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

[Sequences & Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

17 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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18 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

[Vector Calculus · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

[AP Statistics Exam Format & Study Design · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

[Unit 1 · Exploring One-Variable & Two-Variable Data · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value \(w = 10 \). If the gradient of the loss is \(\frac{\partial L}{\partial w} = 4 \) and the learning rate is \(\eta = 0.1 \), find the updated weight \(w' \).

a. \(10.40 \)

b. **\(9.60 \)** ✓

c. \(6.00 \)

d. \(9.90 \)

Using the gradient descent update rule: \(w' = w - \eta \frac{\partial L}{\partial w} = 10 - (0.1)(4) = 10 - 0.40 = 9.60 \).

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22 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the scalar loss function \(L(w) = 2w^2 - 8w + 5 \). Find the analytical gradient \(\nabla L(w) \) evaluated at \(w = 2 \).

a. \(\nabla L(2) = 16 \)

b. \(\nabla L(2) = 8 \)

c. **\(\nabla L(2) = 0 \)** ✓

d. \(\nabla L(2) = -4 \)

Differentiate with respect to \(w \): \(\frac{dL}{dw} = 2(2)w - 8 = 4w - 8 \). At \(w = 2 \): \(\nabla L = 4(2) - 8 = 8 - 8 = 0 \).

23 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A 2D loss surface is defined as $L(w_1, w_2) = 2w_1^2 + 3w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [10, -8]^T$

b. $ablaL = [8, 6]^T$

c. $ablaL = [4, -3]^T$

d. $ablaL = [8, -6]^T$ ✓

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(2)w_1 = 4(2) = 8$; $\frac{\partial L}{\partial w_2} = 2(3)w_2 = 6(-1) = -6$. Therefore, $ablaL = [8, -6]^T$.

24 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In training a language model token predictor, a binary token occurs $k = 3$ times across $n = 10$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

a. $\hat{p}_{\text{MLE}} = 0.300$ ✓

b. $\hat{p}_{\text{MLE}} = 0.450$

c. $\hat{p}_{\text{MLE}} = 0.150$

d. $\hat{p}_{\text{MLE}} = 0.700$

For a Bernoulli likelihood $L(p) = p^k(1-p)^{n-k}$, the log-likelihood is $\ln L = k\ln p + (n-k)\ln(1-p)$. Setting derivative to zero: $\frac{k}{p} - \frac{n-k}{1-p} = 0 \Rightarrow \hat{p} = \frac{k}{n} = \frac{3}{10} = 0.300$.

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25 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

a. 0.0450 ✓

b. 0.9500

c. 0.8500

d. 0.0050

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.05 \times 0.90 = 0.0450$.

26 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

For a bivariate feature dataset, $\text{Cov}(x, y) = 12$, $\text{Var}(x) = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

a. $\hat{\beta}_1 = 0.3333333333333333,\ \hat{\beta}_0 = 10$

b. $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 10$ ✓

c. $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$

d. $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 5$

The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{12}{4} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 25 - (3)(5) = 25 - 15 = 10$.

27 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model?

a. 75.0% ✓

b. 60.0%

c. 25.0%

d. 85.7%

Precision is the fraction of predicted positive instances that are truly positive:
$\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30 + 10} = \frac{30}{40} = 75.0\%$.

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28 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.

a. 81.0% ✓

b. 81.8%

c. 52.4%

d. 13

Accuracy is the ratio of correct predictions to total samples:
$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{45 + 40}{105} = \frac{85}{105} = 81.0\%$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$ ✓

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

The coefficient of determination measures the proportion of variance explained:
$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{20}{100} = 1 - 0.20 = 0.80$.

30 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

a. 0.140

b. 0.950

c. 0.279 ✓

d. 0.391

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.02}{0.95(0.02) + 0.05(0.98)} = \frac{0.0190}{0.0680} = 0.279$.

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31 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1?

a. 62.0% ✓

b. 86.0%

c. 24.0%

d. 72.1%

Total variance is $\sum \lambda_i = 62 + 24 + 14 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{62}{100} = 62.0\%$.

32 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{50}$ onto its top 6 principal components. What is the dimension of each transformed data point?

a. 25

b. 50

c. 44

d. 6 ✓

PCA projection onto k eigenvectors transforms each sample vector from $\mathbb{R}^D$ to $\mathbb{R}^k$. Here $k = 6$, so the reduced dimension is 6.

33 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?

a. $\mu = 25,\ \sigma^2 = 36$ ✓

b. $\mu = 36,\ \sigma^2 = 25$

c. $\mu = 25,\ \sigma^2 = 6$

d. $\mu = 6,\ \sigma^2 = 36$

The notation $\mathcal{N}(\mu, \sigma^2)$ directly specifies the mean $\mu = 25$ and variance $\sigma^2 = 36$ (with standard deviation $\sigma = 6$).

34 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

a. 0.429

b. 0.353

c. -2.347

d. -0.847 ✓

Odds are $\frac{p}{1-p} = \frac{0.30}{0.70} = 0.4286$. Taking natural logarithm: $\ln(0.4286) = -0.847$.

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35 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. 46 ✓

b. 49

c. 42

d. 28

Substitute $x = 4$ into the regression equation: $\hat{y} = 7(4) + 18 = 28 + 18 = 46$.

36 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.10$. What is the probability of the complement class $P(egC_1)$?

a. 0.90 ✓

b. 0.95

c. 0.05

d. 1.10

By the complement rule of probability: $P(egC_1) = 1 - P(C_1) = 1 - 0.10 = 0.90$.

37 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

Gradient descent step: $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf(\mathbf{x}_n)$. For $f = x^2 + y^2$ at $(1, 1)$, $\alpha = 0.5$:

a. $(0.5, 0.5)$

b. $(0, 0)$ ✓

c. $(-1, -1)$

d. $(1.5, 1.5)$

$ablaf = (2, 2)$; step: $(1, 1) - 0.5(2, 2) = (0, 0)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Gradient descent update $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf$ converges to:

a. **a local min for smooth** f ✓

b. global min always

c. a saddle point

d. the origin

Gradient descent: local min for smooth f.

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39 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.

a. **(2, 3)** ✓

b. (-2, -3)

c. (4, 6)

d. (0, 0)

To find the critical point(s) of a function of two variables, we need to find where the gradient is zero. The gradient is given by $ablaf = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$. First, we compute the partial derivative with respect to x: $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^2 - 4x - 6y + 13) = 2x - 4$. Next, we compute the partial derivative with respect to y.

40 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.

a. 10

b. 13

c. **26** ✓

d. 52

We are given $z = f(x, y) = x^2 + y^2$, with $x = 2t$ and $y = 3t$. We need to find $\frac{dz}{dt}$ using the Multivariable Chain Rule: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$. First, find the partial derivatives of z with respect to x and y: $\frac{\partial z}{\partial x} = 2x$ $\frac{\partial z}{\partial y} = 2y$ Next, find the derivatives of x and y with respect to t: $\frac{dx}{dt} = 2$ $\frac{dy}{dt} = 3$ Now, substitute these into.

41 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

To find the critical points of the function $f(x, y) = x^3 + y^3 - 3axy$, we first compute the first partial derivatives and set them to zero: $f_x = 3x^2 - 3ay$ $f_y = 3y^2 - 3ax$ Setting these to zero, we get the system of equations: 1) $3x^2 - 3ay = 0$ => $x^2 = ay$ 2) $3y^2 - 3ax = 0$ => $y^2 = ax$ We are given that $(x_0, y_0) = (2, 2)$ is a critical point. Substituting these values into the equations: From equation 1: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$. From equation 2: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$. Bo.

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42 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

a. **2** ✓

b. 4

c. 8

d. 10

To find the partial derivative of $f(x, y) = x^2 + 2y^2$ with respect to x, we treat y as a constant and differentiate with respect to x. The derivative of x^2 with respect to x is $2x$, and the derivative of $2y^2$ (a constant with respect to x) is 0. Therefore, $\frac{\partial f}{\partial x} = 2x$. Now, we evaluate this partial derivative at the point $(1, 2)$. Substituting $x = 1$ into the expression for $\frac{\partial f}{\partial x}$: $\frac{\partial f}{\partial x}$ at $(1, 2) = 2 \times 1 = 2$. Th.

43 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

a. **2** ✓

b. 4

c. 6

d. 0

First, we find the first partial derivative of $f(x, y)$ with respect to x: $f_x(x, y) = \frac{\partial}{\partial x}(x^2 + y^2) = 2x$. Next, we find the second partial derivative of $f(x, y)$ with respect to x (which is f_{xx}):
$f_{xx}(x, y) = \frac{\partial}{\partial x}(f_x(x, y)) = \frac{\partial}{\partial x}(2x) = 2$. Finally, we evaluate $f_{xx}(x, y)$ at the point $(2, 3)$. Since $f_{xx}(x, y)$ is a constant, its value at any point is 2. Therefore, $f_{xx}(2, 3) = 2$.

44 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of f with respect to y, denoted as $\frac{\partial f}{\partial y}$?

a. $2x - 3y$

b. $2x - 3xy + 4y$

c. $-3x + 4y$ ✓

d. $2x - 3y + 4y$

To find the partial derivative of $f(x, y) = x^2 - 3xy + 2y^2$ with respect to y, we treat x as a constant. \\ \\ The derivative of x^2 with respect to y is 0, as x is considered constant. \\ \\ The derivative of $-3xy$ with respect to y is $-3x \cdot 1 = -3x$. \\ \\ The derivative of $2y^2$ with respect to y is $2 \cdot 2y = 4y$. \\ \\ Therefore, the partial derivative $\frac{\partial f}{\partial y}$ is $0 - 3x + 4y = -3x + 4y$.

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45 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

a. 6

b. **12** ✓

c. -6

d. 15

To find $f_x(2, -3)$, we first compute the partial derivative of $f(x, y)$ with respect to x. Treating y as a constant, we get: $f_x(x, y) = \frac{\partial}{\partial x}(3x^2 - 5y) = 6x$. Now, we evaluate f_x at the point $(2, -3)$: $f_x(2, -3) = 6(2) = 12$.

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